Here are the complete, formal notes on the video "Design Matrices For Linear Models, Clearly Explained!!!".

Subject: General Linear Models (Part 3): Design Matrices

Source: StatQuest with Josh Starmer http://www.youtube.com/watch?v=CqLGvwi-5Pc

I. The "Standard" Design Matrix for a t-test

This video begins by comparing the simple t-test design matrix from Part 2 with the more common "standard" formulation [00:37].

- **Simple Matrix (from Part 2):**
  - Equation: $Y = (\beta_1 \cdot \text{Mean}_{\text{control}}) + (\beta_2 \cdot \text{Mean}_{\text{mutant}}) + \epsilon$
  - Matrix X: Each column is an "on/off" switch for a group's mean.

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

- **Standard Matrix (Interceptor Model):**
  - This model re-parameterizes the equation to use an **intercept** (baseline) and an **offset** (difference) [00:45].
  - Equation: $Y = (\beta_0) + (\beta_1 \cdot \text{Difference}_{\text{mutant\_vs\_control}}) + \epsilon$
  - Matrix X:
    - Column 1 (Intercept): All 1s. This is the baseline, set as $\text{Mean}_{\text{control}}$.
    - Column 2 (Offset): 0 for controls, 1 for mutants. This "turns on" the β_1 term only for the mutant group.

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$$

  - **Interpretation:**
    - For Controls (Y_{control}): $Y = \beta_0(1) + \beta_1(0) = \beta_0 = \text{Mean}_{\text{control}}$
    - For Mutants (Y_{mutant}): $Y = \beta_0(1) + \beta_1(1) = \beta_0 + \beta_1 = \text{Mean}_{\text{mutant}}$
    - Here, the parameter β_1 explicitly represents the *difference* between the two group means.
- **Equivalency:**
  - Both models have the same number of parameters ($p_{\text{fit}} = 2$) [02:06].
  - They produce the exact same fitted values ($\hat{y}$) for each data point.
  - Therefore, the Sum of Squared Residuals (SS_{fit}) is identical.
  - This results in the **exact same F-statistic and p-value** [02:24]. The "standard" model is more common because it is more easily extended, as seen in regression.

II. The General Linear Model Framework

The design matrix is a general method for structuring an equation that can be solved with least squares. The values in the matrix are simply **multipliers** for the parameters in the model equation [03:01].

- **General Equation:** $Y = X\beta + \epsilon$
  - Y: The vector of observed outcomes (dependent variable).
  - X: The $n \times p$ design matrix (where n = # of observations, p = # of parameters).
  - β: The vector of parameters to be estimated.

- ϵ: The vector of residuals (error).

III. The Design Matrix for Linear Regression

The key insight is that a design matrix is not limited to 0s and 1s (categorical data). It can also incorporate continuous data [03:47].

- **Model:** Simple Linear Regression
- **Equation:** $Y_i = \beta_0 + (\beta_1 \cdot X_i) + \epsilon_i$
  - β_0: The y-intercept.
  - β_1: The slope.
  - X_i: The continuous predictor value for observation i.
- **Design Matrix X:**
  - Column 1 (Intercept, β_0): A column of all 1s (to "turn on" the intercept for every observation) [06:13].
  - Column 2 (Slope, β_1): The vector of x-axis values (the continuous data itself) [04:04].

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$$

- **Mechanism:** When the i-th row of the matrix $[1, x_i]$ is applied to the parameter vector $[\beta_0, \beta_1]$, it calculates the fitted value $\hat{y}_i$:

$$\hat{y}_i = (1 \cdot \beta_0) + (x_i \cdot \beta_1)$$

IV. Combining Categorical and Continuous Data (ANCOVA)

The true power of the design matrix is its ability to flexibly combine different types of predictors into a single model [06:30]. This example tests if two groups (Control, Mutant) have different regression lines for mouse weight vs. size.

- **Model:** Analysis of Covariance (ANCOVA), assuming parallel slopes.
- **Goal:** Fit two parallel lines and test if the intercept offset is significant [08:05].
- **Equation:** $Y_i = \beta_0 + (\beta_1 \cdot \text{Mutant\_Offset}_i) + (\beta_2 \cdot \text{Weight}_i) + \epsilon_i$
  - β_0: The y-intercept for the baseline group (Controls).
  - β_1: The *difference* in intercepts for the Mutant group (the vertical offset).
  - β_2: The single, common slope for both groups.
- **Design Matrix X ($p = 3$):**
  - Column 1 (Intercept, β_0): All 1s.
  - Column 2 (Group Offset, β_1): 0 for Controls, 1 for Mutants.
  - Column 3 (Covariate/Slope, β_2): The continuous x-values (Weight).

$$X = \begin{bmatrix} 1 & 0 & \text{weight}_1 \\ 1 & 0 & \text{weight}_2 \\ 1 & 1 & \text{weight}_3 \\ 1 & 1 & \text{weight}_4 \end{bmatrix} \begin{array}{l} (\text{Control}) \\ (\text{Control}) \\ (\text{Mutant}) \\ (\text{Mutant}) \end{array}$$

- **Interpretation:**
  - For Controls: $\hat{y} = \beta_0 + (\beta_1 \cdot 0) + (\beta_2 \cdot \text{Weight}) = \beta_0 + (\beta_2 \cdot \text{Weight})$ (Line 1)
  - For Mutants: $\hat{y} = \beta_0 + (\beta_1 \cdot 1) + (\beta_2 \cdot \text{Weight}) = (\beta_0 + \beta_1) + (\beta_2 \cdot \text{Weight})$ (Line 2)

V. Generalized Hypothesis Testing (Model Comparison)

The design matrix allows for a generalized F-test to compare any two **nested models** (where one "Simple Model" is a subset of a "Fancy Model") [09:57].

- **Generalized F-Statistic:**

$$F = \frac{(SS_{\text{simple}} - SS_{\text{fancy}})/(p_{\text{fancy}} - p_{\text{simple}})}{SS_{\text{fancy}}/(n - p_{\text{fancy}})}$$

  - SS_{simple}: Sum of Squares Residuals from the simpler model.
  - SS_{fancy}: Sum of Squares Residuals from the more complex model.
  - p_{simple}: Number of parameters in the simpler model.
  - p_{fancy}: Number of parameters in the complex model.
  - n: Sample size.

The p-value from this F-statistic answers: **"Is the *improvement* in fit from adding the extra parameters to the fancy model statistically significant?"** [12:34]

- **Example 1: ANCOVA vs. Simple Regression** [11:13]
  - **Fancy Model:** $Y = \beta_0 + \beta_1(\text{Offset}) + \beta_2(\text{Weight})$ ($p_{\text{fancy}} = 3$)
  - **Simple Model:** $Y = \beta_0 + \beta_2(\text{Weight})$ ($p_{\text{simple}} = 2$)
  - **Question:** Is adding the group offset (i.e., using two lines) significantly better than just one regression line?
- **Example 2: ANCOVA vs. t-test** [11:59]
  - **Fancy Model:** $Y = \beta_0 + \beta_1(\text{Offset}) + \beta_2(\text{Weight})$ ($p_{\text{fancy}} = 3$)
  - **Simple Model:** $Y = \beta_0 + \beta_1(\text{Offset})$ ($p_{\text{simple}} = 2$)
  - **Question:** Is adding the weight covariate (the slope) significantly better than just comparing the group means?

VI. Application: Batch Effect Correction

This framework can solve complex real-world problems, such as combining data from different experiments [12:46].

- **Problem:** Lab A (Batch 1) and Lab B (Batch 2) both test Control vs. Mutant. Lab B's measurements are systematically lower (a "batch effect") [12:54].
- **Goal:** Test for the mutant effect *after* compensating for the batch effect.
- **"Fancy" Model Equation:**

$$Y = \beta_0 + (\beta_1 \cdot \text{LabB\_Offset}) + (\beta_2 \cdot \text{Mutant\_Effect}) + \epsilon$$

- **Design Matrix X ($p = 3$):**
  - β_0: Intercept (Mean of Control in Lab A). Column 1 is all 1s.
  - β_1: Batch Effect. Column 2 is 0 for Lab A, 1 for Lab B.
  - β_2: Treatment Effect. Column 3 is 0 for Controls, 1 for Mutants.
- **"Simple" Model (Null Hypothesis):**
  - We create a model that accounts for the batch effect but *not* the treatment effect.
  - Equation: $Y = \beta_0 + (\beta_1 \cdot \text{LabB\_Offset}) + \epsilon$ ($p_{\text{simple}} = 2$)
- **Hypothesis Test:**
  - We compare the SS_{fancy} (from the 3-parameter model) to the SS_{simple} (from the 2-parameter model).
  - A significant p-value indicates that the β_2 (Mutant_Effect) term is statistically significant, *even after* "soaking up" the variance caused by the batch effect [14:05].

http://googleusercontent.com/youtube_content/2